Unit 8 Exercises

4SC000 Optimal Control and Reinforcement Learning

Duarte Antunes

A runnable Python companion to these exercises – LQR with input constraints, MPC for reference tracking, structured (SLS) synthesis, and MPC-based rollout – is available in `code/unit8/` (see its `README.md`).

Core exercises

Problem 8.1 Consider the switched linear system $x_{k+1} = A_{\sigma_k} x_k$, $k \in \mathbb{N}_0$, where $\sigma_k \in \{0, 1\}$ is the control input, and

$$A_0 = \begin{bmatrix} 0.8 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}, \qquad A_1 = \begin{bmatrix} 1 & 0.1 \\ 0 & 0.5 \end{bmatrix}.$$

Suppose that we wish to minimize $\|x_h\|^2$ for some $h \in \mathbb{N}$.

(i) Provide the optimal policy for σ_0 and σ_1 when $h = 2$.
(ii) Consider the base policy $\sigma_k = 1$, $\forall k \in \{0, \dots, h-1\}$. Considering this base policy, derive the rollout policy with horizon $H = 1$ for σ_0, σ_1, σ_2 when $h = 3$.

Solution

(i) This is a standard stage decision problem with $g_0 = g_1 = 0$ and terminal cost $g_2(x_2) = x_2^\top x_2$. Applying DP backward:

$$J_2(x_2) = x_2^\top x_2, \qquad J_1(x_1) = \min_{\sigma_1 \in \{0,1\}} x_1^\top A_{\sigma_1}^\top A_{\sigma_1} x_1 = \min \left\{ x_1^\top \begin{bmatrix} 0.68 & 0.24 \\ 0.24 & 0.65 \end{bmatrix} x_1,\ x_1^\top \begin{bmatrix} 1 & 0.1 \\ 0.1 & 0.26 \end{bmatrix} x_1 \right\}.$$

The optimal policy at stage $k = 1$ is

$$\sigma_1 = \begin{cases} 0 & \text{if } x_1^\top \begin{bmatrix} 0.68 & 0.24 \\ 0.24 & 0.65 \end{bmatrix} x_1 < x_1^\top \begin{bmatrix} 1 & 0.1 \\ 0.1 & 0.26 \end{bmatrix} x_1 \\ 1 & \text{otherwise,} \end{cases}$$

equivalently $\sigma_1 = 0$ if $x_1^\top \begin{bmatrix} -0.32 & 0.14 \\ 0.14 & 0.39 \end{bmatrix} x_1 < 0$, a conic partition of the state space (the policy is not unique on the boundary). Writing $x_1 = (y, z)$, the region where $\sigma_1 = 0$ is $\{(y, z) : -0.32y^2 + 0.28zy + 0.39z^2 < 0\}$; setting $y = \alpha z$ gives $-0.32\alpha^2 + 0.28\alpha + 0.39 = 0$, i.e. $\alpha = 1.625$ or $\alpha = -0.75$, so the boundary consists of the two lines $y = 1.625z$ and $y = -0.75z$.
At stage $k = 0$,

$$J_1(x_1) = \min\{\|A_0 x_1\|^2, \|A_1 x_1\|^2\}, \qquad J_0(x_0) = \min_{\sigma_0 \in \{0,1\}} \{\|A_0 A_{\sigma_0} x_0\|^2, \|A_1 A_{\sigma_0} x_0\|^2\} = \min\{x_0^\top Z_i x_0\}_{i=1}^4,$$

where $Z_1 = (A_0 A_0)^\top A_0 A_0$, $Z_2 = (A_0 A_1)^\top A_0 A_1$, $Z_3 = (A_1 A_0)^\top A_1 A_0$, $Z_4 = (A_1 A_1)^\top A_1 A_1$, and

$$\sigma_0 = \begin{cases} 0 & \text{if } x_0^\top Z_1 x_0 < x_0^\top Z_i x_0 \text{ for } i \in \{2,3,4\}, \text{ or } x_0^\top Z_3 x_0 < x_0^\top Z_i x_0 \text{ for } i \in \{1,2,4\} \\ 1 & \text{otherwise.} \end{cases}$$

(ii) For the $H = 1$ rollout with base policy $\sigma_k = 1$ for all k, the approximate cost-to-go replaces the exact tail cost by the base policy's cost from H steps ahead: for $h = 3$ the tail after one rolled-out decision is $x_3 = A_1 A_1 A_{\sigma_0} x_0$, so at stage 0 the rollout policy minimizes $\|A_1 A_1 A_{\sigma_0} x_0\|^2$ over σ_0:

$$\sigma_0 = \begin{cases} 0 & \text{if } \|A_1 A_1 A_0 x_0\|^2 < \|A_1 A_1 A_1 x_0\|^2 \\ 1 & \text{otherwise.} \end{cases}$$

By the same reasoning, applied one and two steps later,

$$\sigma_1 = \begin{cases} 0 & \text{if } \|A_1 A_0 x_1\|^2 < \|A_1 A_1 x_1\|^2 \\ 1 & \text{otherwise,} \end{cases} \qquad \sigma_2 = \begin{cases} 0 & \text{if } \|A_0 x_2\|^2 < \|A_1 x_2\|^2 \\ 1 & \text{otherwise.} \end{cases}$$

Problem 8.2 Consider the continuous-time switched linear system $\dot{x}(t) = B_{\sigma(t)} x(t)$, $x(t) = \begin{bmatrix} y(t) & z(t) \end{bmatrix}^\top$, with

$$B_1 = \begin{bmatrix} -(1-\beta) & 1 \\ -9 & -(1-\beta) \end{bmatrix}, \qquad B_2 = \begin{bmatrix} -(1-\beta) & 9 \\ -1 & -(1-\beta) \end{bmatrix}.$$

The system is controlled digitally with sampling period $\tau = 0.1$: $x_{k+1} = A_{\sigma_k} x_k$, $A_1 = e^{B_1 \tau}$, $A_2 = e^{B_2 \tau}$.

(i) Suppose $\beta = 1$. (a) Provide a policy for $\sigma(t)$ stabilizing the continuous-time system. (b) Provide a policy for σ_k stabilizing the discrete-time system.

(ii) Consider the cost $\sum_{k=0}^{\infty} x_k^\top x_k$. (a) Derive the policy obtained by MPC with horizon $H = 2$ (minimizing $x_k^\top x_k + x_{k+1}^\top x_{k+1}$ at each k), and describe the switching regions for $\beta = 1$ and $\beta = 0.9$. Does this policy stabilize the discrete-time system? (b) For $\beta = 0.9$, derive the rollout policy with horizon $H = 1$ using base policy $\sigma_k = 1$ for every $k \geq 0$. Is this base policy usable when $\beta = 1$?

i Solution

(i)(a) With $\beta = 1$, $B_1 = \begin{bmatrix} 0 & 1 \\ -9 & 0 \end{bmatrix}$ and $B_2 = \begin{bmatrix} 0 & 9 \\ -1 & 0 \end{bmatrix}$: mode 1 gives $\dot{y} = z$, $\dot{z} = -9y$, and mode 2 gives $\dot{y} = 9z$, $\dot{z} = -y$ – both undamped oscillators (eigenvalues $\pm 3i$), individually marginal rather than stabilizing. Using $V(x) = y^2 + z^2$ as a candidate Lyapunov function, $\dot{V} = 2y\dot{y} + 2z\dot{z}$ evaluates to $\dot{V} = -16yz$ under mode 1 and $\dot{V} = 16yz$ under mode 2. Switching to whichever mode makes $\dot{V}$ negative therefore stabilizes the system:

$$\sigma(t) = \begin{cases} 1 & \text{if } y(t)z(t) \geq 0 \\ 2 & \text{if } y(t)z(t) < 0. \end{cases}$$

(The source figure for this problem could not be reproduced here; this policy is verified directly by the sign of $\dot{V}$ above rather than read off a plot.)

(i)(b) Since $\tau = 0.1$ is small compared with the oscillation period of either mode ($2\pi/3 \approx 2.1$), the same state-based rule stabilizes the sampled-data system: $\sigma_k = 1$ if $y_k z_k \geq 0$, $\sigma_k = 2$ if $y_k z_k < 0$.

(ii)(a) At time k, x_k is fixed, so minimizing $x_k^\top x_k + x_{k+1}^\top x_{k+1} = x_k^\top x_k + x_k^\top A_{\sigma_k}^\top A_{\sigma_k} x_k$ over σ_k is equivalent to minimizing $x_k^\top A_{\sigma_k}^\top A_{\sigma_k} x_k$ alone:

$$\sigma_k = \begin{cases} 1 & \text{if } x_k^\top (A_1^\top A_1 - A_2^\top A_2) x_k < 0 \\ 2 & \text{otherwise.} \end{cases}$$

Computing A_1, A_2 numerically: for $\beta = 1$, $A_1^\top A_1 - A_2^\top A_2 = \begin{bmatrix} 0.776 & -1.506 \\ -1.506 & -0.776 \end{bmatrix}$, an indefinite matrix whose eigenvalues are ± 1.69; the switching region boundary is the pair of lines $y = 4.12z$ and $y = -0.243z$ through the origin. For $\beta = 0.9$ the same computation gives $A_1^\top A_1 - A_2^\top A_2 = \begin{bmatrix} 0.761 & -1.476 \\ -1.476 & -0.761 \end{bmatrix}$, with essentially the same boundary lines. What differs between the two cases is the open-loop behaviour of each mode alone: at $\beta = 1$, $|\text{eig}(A_1)| = |\text{eig}(A_2)| = 1$ (marginally stable/undamped, matching (i)); at $\beta = 0.9$, $|\text{eig}(A_1)| = |\text{eig}(A_2)| = 0.99 < 1$ (each mode is already, if barely, asymptotically stable on its own).

Consequently, at $\beta = 1$ the one-step-lookahead MPC policy is exactly the greedy, myopic version of the stabilizing switching rule from (i)(b) – since $H = 2$ only looks one step ahead, it need not accumulate the sustained sign pattern in $y_k z_k$ that true stabilization requires, so MPC does not guarantee stability of the switched system in general (this is the standard caveat that MPC's finite lookahead is not automatically stabilizing). At $\beta = 0.9$, stability holds regardless of which mode MPC selects at each step, since both individual modes already contract; the switching decision only affects the transient, not asymptotic stability.

(ii)(b) For $\beta = 0.9$, A_1 is stable ($|\text{eig}(A_1)| = 0.99 < 1$), so the infinite-horizon cost of the base policy $\sigma_k = 1$ for all k is $\sum_{k=0}^{\infty} x_k^\top x_k = x_0^\top P x_0$, where P solves the Lyapunov equation $A_1^\top P A_1 - P = -I$ (the general result proved in Unit 5, Problem 5.4). Numerically, $P = \begin{bmatrix} 250.3 & -2.15 \\ -2.15 & 28.3 \end{bmatrix}$ (the large entries reflect that A_1 is close to the stability boundary). The $H = 1$ rollout policy is then

$$\sigma_k = \begin{cases} 1 & \text{if } x_k^\top (A_1^\top P A_1 - A_2^\top P A_2) x_k < 0 \\ 2 & \text{otherwise.} \end{cases}$$

This base policy cannot be used when $\beta = 1$: there, A_1 is only marginally stable ($|\text{eig}(A_1)| = 1$), so the Lyapunov equation $A_1^\top P A_1 - P = -I$ has no positive definite solution, and the cost of applying $\sigma_k = 1$ forever is unbounded – the base policy is not stabilizing and so cannot anchor a rollout.

Problem 8.3 Consider the system $x_{k+1} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_k$, $u_k \in \mathbb{R}$, $x_k \in \mathbb{R}^2$, with two available stabilizing controllers $u_k = \begin{bmatrix} -\frac{1}{2} & -1 \end{bmatrix} x_k$ and $u_k = \begin{bmatrix} -\frac{1}{3} & -\frac{6}{5} \end{bmatrix} x_k$. The goal is to switch between the two to minimize $\sum_{k=0}^{\infty} \|x_k\|^2$.

(i) Formulate the problem as optimal control of a switched linear system $x_{k+1} = A_{\sigma_k} x_k$, $\sigma_k \in \{1, 2\}$, with cost $\sum_{k=0}^{\infty} x_k^\top Q x_k$ for some positive definite Q. Provide A_1, A_2, Q.

(ii) Using the Lyapunov equation, show that the cost of the base policy $\sigma_k = 1$, $\forall k$, takes the form $x_0^\top \bar{P} x_0$. Provide $\bar{P}$.

(iii) Considering the same base policy, derive the rollout policy with horizon $H = 1$ for $\{\sigma_k\}_{k=0}^{\infty}$.

(iv) Let $\hat{J}(x_0)$ denote the cost when the $H = 1$ rollout policy is applied for a given x_0. It can be shown that $\hat{J}(x_0) \leq x_0^\top \bar{P} x_0$ for every x_0. Use this to establish that the rollout policy asymptotically stabilizes the system.

i Solution

(i) Substituting each controller into the dynamics, $A_1 = A + BK_1 = \begin{bmatrix} 1 & 1 \\ -\frac{1}{2} & 0 \end{bmatrix}$ and $A_2 = A + BK_2 = \begin{bmatrix} 1 & 1 \\ -\frac{1}{3} & -\frac{1}{5} \end{bmatrix}$, and since the cost is $\sum \|x_k\|^2 = \sum x_k^\top x_k$, $Q = I$.

(ii) A_1 is asymptotically stable ($\text{eig}(A_1) = 0.5 \pm 0.5i$, $|\text{eig}(A_1)| = 1/\sqrt{2}$), so by the Lyapunov-equation result of Unit 5, Problem 5.4, the cost of applying $\sigma_k = 1$ forever is $x_0^\top \bar{P} x_0$ where $\bar{P}$ solves $A_1^\top \bar{P} A_1 - \bar{P} = -Q$. Solving exactly gives

$$\bar{P} = \begin{bmatrix} 3 & 2 \\ 2 & 4 \end{bmatrix}.$$

(iii) The $H = 1$ rollout policy at state x_k minimizes $x_k^\top Q x_k + x_{k+1}^\top \bar{P} x_{k+1} = x_k^\top Q x_k + x_k^\top A_{\sigma_k}^\top \bar{P} A_{\sigma_k} x_k$ over $\sigma_k \in \{1, 2\}$; since the Q-term does not depend on σ_k, this is equivalent to minimizing $x_k^\top A_{\sigma_k}^\top \bar{P} A_{\sigma_k} x_k$. Computing exactly, $A_1^\top \bar{P} A_1 = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix}$ and $A_2^\top \bar{P} A_2 = \begin{bmatrix} 19/9 & 11/5 \\ 11/5 & 59/25 \end{bmatrix}$, so

$$A_1^\top \bar{P} A_1 - A_2^\top \bar{P} A_2 = \begin{bmatrix} -1/9 & -1/5 \\ -1/5 & 16/25 \end{bmatrix},$$

and, writing $x_k = (y, z)$, the (stationary, since the base-policy cost-to-go $\bar{P}$ is the same at every stage) rollout policy is

$$\sigma_k = \begin{cases} 1 & \text{if } -\frac{1}{9}y^2 - \frac{2}{5}yz + \frac{16}{25}z^2 < 0 \\ 2 & \text{otherwise.} \end{cases}$$

(iv) *(This part is not covered by the source solutions; the argument below is derived independently.)* Because the rollout policy is stationary and state-feedback, applying it from any state x gives a well-defined cost $\hat{J}(x) = \sum_{j \geq 0} x_j^\top Q x_j$ (trajectory started at $x_0 = x$), which decomposes recursively as $\hat{J}(x_k) = x_k^\top Q x_k + \hat{J}(x_{k+1})$ along the closed-loop trajectory generated by the policy. Combined with the given bound $\hat{J}(x) \leq x^\top \bar{P} x$ (valid for every state, in particular for every x_k visited along the trajectory), $\hat{J}$ is a valid candidate Lyapunov function for the closed-loop system: it is positive definite ($\hat{J}(x) \geq 0$ since $Q \geq 0$, with equality only at $x = 0$), finite everywhere (bounded above by $x^\top \bar{P} x < \infty$), and satisfies

$$\hat{J}(x_k) - \hat{J}(x_{k+1}) = x_k^\top Q x_k \geq 0,$$

with strict inequality whenever $x_k \neq 0$ (since $Q = I > 0$). So $\hat{J}(x_k)$ is non-increasing along the trajectory and strictly decreasing unless $x_k = 0$; being bounded below by 0, it converges, which forces the increments $x_k^\top Q x_k \to 0$, and since $Q > 0$ this means $\|x_k\| \to 0$. Hence the $H = 1$ rollout policy asymptotically stabilizes the system.

Additional exercises

Problem 8.4 Consider $\begin{bmatrix} \dot{y}(t) \\ \dot{z}(t) \end{bmatrix} = \begin{bmatrix} \cos(\theta(t)) \\ \sin(\theta(t)) \end{bmatrix}$, where $\theta(t) \in [-\pi, \pi)$ is the control input and $x(t) = \begin{bmatrix} y(t) & z(t) \end{bmatrix}^\top$ is the state. The input is implemented digitally, $\theta(t) = \theta_k$ for $t \in [k\tau, (k+1)\tau)$, giving the Euler discretization $x_{k+1} = x_k + \tau \begin{bmatrix} \cos(\theta_k) \\ \sin(\theta_k) \end{bmatrix}$. The goal is to minimize $\sum_{k=0}^{\infty} \|x_k\|_Q^2$ for positive definite Q.

(i) Show that the MPC policy with horizon $H = 2$ is $\theta_k(x_k) = \mu(x_k)$, where

$$\mu(x) = \begin{cases} \arctan\left(\frac{v_2(x)}{v_1(x)}\right) & \text{if } v_1(x) \neq 0 \\ \frac{\pi}{2} & \text{if } v_2(x) \geq 0,\ v_1(x) = 0 \\ -\frac{\pi}{2} & \text{if } v_2(x) < 0,\ v_1(x) = 0, \end{cases}$$

and $v(x) = (v_1(x), v_2(x))$ solves $\min_{v \in \mathbb{R}^2} \|x + \tau v\|_Q^2$ s.t. $\|v\|^2 = 1$. (ii) Provide the optimal control input at stage 0 when $\tau = 0.1$, $Q = I$, $x_0 = \begin{bmatrix} 1 & 1 \end{bmatrix}^\top$.

Solution

(i) At time ℓ, the $H = 2$ MPC problem is $\min_{\theta_\ell} \|x_\ell\|_Q^2 + \|x_{\ell+1}\|_Q^2$; since x_ℓ is fixed, this is equivalent to $\min_{\theta_\ell} \|x_\ell + \tau[\cos\theta_\ell, \sin\theta_\ell]^\top\|_Q^2$. Because $\{(\cos\theta, \sin\theta) : \theta \in [-\pi, \pi)\} = \{v \in \mathbb{R}^2 : \|v\|^2 = 1\}$, this is equivalent to the constrained problem $\min_v \|x_\ell + \tau v\|_Q^2$ s.t. $\|v\|^2 = 1$, with $\cos\theta = v_1$, $\sin\theta = v_2$ recovering θ via the four-quadrant relation stated (using arctan on the ratio when $v_1 \neq 0$, and the two vertical-axis cases otherwise).
(ii) With $\tau = 0.1$, $Q = I$, $x_0 = [1 \;\; 1]^\top$, the problem is $\min_{v_1,v_2}(1+0.1v_1)^2+(1+0.1v_2)^2$ s.t. $v_1^2+v_2^2 = 1$. The Lagrangian $L = (1 + 0.1v_1)^2 + (1 + 0.1v_2)^2 + \lambda(v_1^2 + v_2^2 - 1)$ gives, from $\partial L/\partial v_1 = \partial L/\partial v_2 = 0$, $v_1 = v_2 = \frac{-1}{\lambda+0.1}$; substituting into the constraint gives $\lambda = -0.1 \pm \sqrt{2}$. The two stationary points are

$$(v_1, v_2) = \left(-\tfrac{1}{\sqrt{2}}, -\tfrac{1}{\sqrt{2}}\right) \;\Rightarrow\; \theta = \tfrac{5\pi}{4} \text{ (minimizer)}, \qquad (v_1, v_2) = \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}\right) \;\Rightarrow\; \theta = \tfrac{\pi}{4} \text{ (maximizer)}.$$

The optimal control input at stage 0 is therefore $\theta_0 = 5\pi/4$ (pointing away from the origin's opposite side, i.e. steering back toward the origin from $x_0 = [1 \;\; 1]^\top$).